

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	24.8
Total 1 mark			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains the correct mean	AO1.1b	B1	1.38
(i)				
(ii)	Uses the correct formula for standard deviation	AO1.1a	M1	$\sqrt{\frac{261.8}{120} - 1.38^2}$
	Obtains the correct standard deviation	AO1.1b	A1	0.526 to 0.529
(b)	Uses the model to calculate a normal probability	AO3.4	M1	0.5417 to 0.5428
(i)	Obtains correct probability	AO1.1b	A1	
(ii)	Recalls correct value of 0	AO1.2	B1	0
(c)	Calculates the value of mean – 3 × standard deviations	AO1.1b	M1	–0.1998 to –0.207
	Concludes that model might be inappropriate as the value is less than 0	AO2.2b	A1	This is less than 0 and so model might not be appropriate
(d)	Standardises appropriately and formalises a probability statement PI by fully correct equation	AO3.1b	M1	$P\left(Z > \frac{0.75 - \mu}{0.21}\right) = 0.1$ $z \text{ value} = 1.2816$ $\frac{0.75 - \mu}{0.21} = 1.2816$ $\mu = 0.481$
	Obtains z value from inverse normal distribution (± 1.2816)	AO1.1a	M1	
	Forms a correct equation using standardised result and z value	AO1.1b	A1	
	Solves the equation to find	AO1.1b	A1	

the correct value of μ			
Total 12 marks			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies likely outlier	AO1.2	B1	East region 2007
(i)				
(ii)	Finds Q_1 and Q_3 for East region FT NW if that was their outlier in part (a)(i)	AO1.1b	B1F	For East region $Q_1 = 0$ $Q_3 = 4$ (For NW $Q_1 = 2$ $Q_3 = 6$)
	Finds IQR for East region FT NW if that was their outlier in part (a)(i)	AO1.1b	B1	$IQR = 4$ E ($IQR = 4$ NW)
	Completes argument that uses formula given in the question together with 'their' values found for Q_1 , Q_3 and IQR and 'their' outlier for (a)(i) to confirm that it is an outlier Award credit here provided 'their' values do confirm 'their' identified value to be an outlier	AO2.1	R1	$4 + 1.5 \times 4 = 10$ $21 > 10$ Hence East 2007 is an outlier
(b)	Explains reason for mean being unrepresentative	AO2.4	E1	Mean would be unrepresentative as it would be affected by the large value outlier.
(c)	Provides explanation	AO2.2b	E1	Allow: Error in data entry; Some event in 2007 led to an increase in dried milk products consumption locally; Disease hit dairy herds.
Total 6 marks				

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Finds mean	AO1.1b	B1	Mean $\bar{X} = \frac{3046.14}{100} = 30.46(14)$
	Finds standard deviation awfw (4.17 – 4.20) Allow $s =$ $\sqrt{\frac{1746.29}{100}} = 4.179$	AO1.1b	B1	Standard deviation $s = \sqrt{\frac{1746.29}{99}} = 4.20$
	(ii) States max range for a normal distribution	AO1.2	M1	$\bar{X} + 3s \approx 43$ $\bar{X} - 3s \approx 18$
	Explains why normal model acceptable	AO2.4	E1	Normal dist model acceptable because data can be regarded as continuous and the range of amounts is within $\bar{X} \pm 3s$
(iii)	Finds probability awfw (0.095 – 0.097)	AO1.1b	B1	$P(X < 25) = 0.0967$
(b)	Finds z from calculator	AO1.1b	B1	$P\left(Z < \frac{30 - 29.55}{\sigma}\right) = 0.55$
	Uses Normal model to form equation	AO3.4	M1	$\frac{30 - 29.55}{\sigma} = 0.1257$
	Finds correct sd Correct accuracy required for this mark Accept 3.5 or 3.7	AO1.1b	A1	$\sigma = \frac{30 - 29.55}{0.1257} = 3.6$
				Total 8 marks

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Q5.

Mean = 62.2 to 62.3

AWFW (62.25)

B1

SD = 17.4 to 17.6 or 16.7 to 16.9

AWFW (17.519 or 16.774)

B1

[2]

Q6.

151.5 cm = 1.515 m $\approx$ 1.5 m

OE

B1

Thus (could be) correct

OE; dependent on B1

Bdep1

$$n \times 26.6 > 10$$

OE; accept any $n \geq 1$

B1

Thus (likely to be) incorrect

OE; dependent on B1

Bdep1

[4]

Q7.

(a) Mean = 29.2
CAO

B2

Mean = 29 to 30
AWFW

$(\Sigma fx = 730)$

(B1)

2

Note If B0 then clear use of $\Sigma fx / 25 \Rightarrow$ M1

$$\frac{(219 \times 5.50) + (73 \times 6.50) + (438 \times 9.5)}{25}$$

(b) Mean =
Allow divisor of 730 $\Rightarrow 8$

M1

$$= \frac{5840}{25} = \underline{\text{£233 to £234}}$$

AWFW; ignore units (£233.60)

A1

2

(c) $29.2 > 28$ **and / or** $(\text{£233 to £234}) < \text{£250}$
Either

B1

so do **not purchase** business

*Dependent on previous A1
and comparison with £250*

Adep1

2

[6]

Q8.

- (a) (i) Mode = 23
CAO

B1

1

- (ii) Median (88th value) = 22
CAO

B1

Upper quartile (132nd value) = 23

Lower quartile (44th value) = 20

CAO; either
May be implied by IQR = 3

B1

Interquartile range = 3

CAO; do **not** award if **seen** to be not based on 23 and 20

B1

3

- (b) Mean = 22.3
CAO; but only award B1 (22.3) if incorrect mid-points
or $\sum fx$ **seen**

B2

Mean = 21 to 23

AWFW

($\sum fx = 3902.5$)

(B1)

Standard deviation = 6.37 or 6.39

AWRT

($s = 6.391$ $\sigma = 6.372$)

B2

Standard deviation = 5 to 7

AWFW

($\sum fx^2 = 94132.25$)

(B1)

SC Only if B0 B0 or B1 B0 then award as follows but only up to a **maximum total part mark of 2**

1 At least 2 correct mid-points 4.5, 14.5, 27, 32, 37, 44.5, 54 seen $\Rightarrow$ M1

2 Clear use of $\sum fx / (175 \text{ or } 174) \Rightarrow$ M1

(B1)

4

(c) Mean = (c's mean from (b)) + $\frac{280}{175} = 22.3 + 1.6$
Adding (1.6 or equivalent) CAO
to (c's mean from (b)) or to (c's new mean)

M1

Mean = **23.9**

F on (c's mean from (b)) or on (c's new mean)

AF1

2

[10]

Q9.

(a) (i) Mode = **253**
 CAO

B1

1

(ii) Median = **252**
 CAO

B1

Upper quartile = **253**

Lower quartile = **250**

CAO; either

May be implied by IQR = 3

B1

Interquartile range = **3**

*CAO; do not award if **seen** to be not based on 253 and 250*

B1

3

- (b) Mean, $\bar{x} = 251 \text{ to } 251.4$
 Award B1 if divisor **seen** not to be 85 but answer in range

$$\text{AWFW} \quad \sum fx = 21352 \quad \bar{x} = 251.2$$

B2

Note:

If B0 then can award

M1 for attempt at $\sum fx \div 85$ **seen**

*Ignore notation and condone incorrect midpoints
 (eg upper or lower limits used)*

Standard deviation, s or $\sigma = 4.21 \text{ to } 4.28$

Award B1 if divisor **seen** not to be 84 or 85 but answer in range

$$\text{AWFW} \quad \sum fx^2 = 5365134$$

$$\sigma = 4.217 \quad s = 4.242$$

B2

4

- (c) Interquartile range (IQR)

Named

B1

Not affected by unknown/large/small/extreme/outlying/227 & 271 values

Or equivalent

Dependent on previous B1

Only negative comments on other measures $\Rightarrow$ Bdep0

Bdep1

OR

More than one named $\Rightarrow$ B0 Bdep0

Range $\Rightarrow$ B0 Bdep0

Standard deviation (s or σ)

Named

(B1)

Uses all data values

Or equivalent

Dependent on previous (B1)

Only negative comments on other measures $\Rightarrow$ Bdep0

(Bdep1)

2

[10]

Q10.

- (a) (i) Mode = **253**
CAO

B1

1

- (ii) Median = **252**
CAO

B1

Upper quartile = **253**

Lower quartile = **250**

CAO; *either*

May be implied by IQR = 3

B1

Interquartile range = **3**

CAO; *do not award if **seen** to be not based on 253 and 250*

B1

3

- (b) (i) Range = $271 - 227 = \mathbf{44}$

CAO; *do not award if **seen** to be not based on 271 and 227*

B1

1

- (ii) Mean, $\bar{x} = \mathbf{251 \text{ to } 251.4}$
*Award B1 if divisor **seen** not to be 85 but answer in range*

$$\text{AWFW } \sum fx = 21352 \quad \bar{x} = 251.2$$

B2

Note:

If B0 then can award

M1 for attempt at $\sum fx \div 85$ **seen**

*Ignore notation and condone incorrect midpoints
(eg upper or lower limits used)*

Standard deviation, s or $\sigma = \mathbf{4.21 \text{ to } 4.28}$

*Award B1 if divisor **seen** not to be 84 or 85 but answer in range*

$$\text{AWFW } \sum fx^2 = 5365134$$

$$\sigma = 4.217 \quad s = 4.242$$

B2

4

- (c) Interquartile range (IQR)
Named

B1

Not affected by unknown/large/small/extreme/outlying/227 & 271 values
Or equivalent
Dependent on previous B1
Only negative comments on other measures $\Rightarrow$ Bdep0

Bdep1

OR

More than one named $\Rightarrow$ B0 Bdep0
Range $\Rightarrow$ B0 Bdep0

Standard deviation (s or σ)
Named

(B1)

Uses all data values
Or equivalent
Dependent on previous (B1)
Only negative comments on other measures $\Rightarrow$ Bdep0
(Bdep1)

2

[11]

Q11.

- (a) Ordering values gives:
May be implied by correct median or correct IQR

(a) 14 15 18 20 25 25 26 27 29 32 34 37 37 (b)
Ignore any reference to a and b

M1

Median = 26
CAO

A1

IQR = 34 – 18 = 16
CAO

A2

Special Case:

Identification that $LQ = 18$ and $UQ = 34$

Both CAO

(A1)

4

- (b) (i) Two values (25 and 37) of mode
 No unique value
 Sparse data
 Many different values
Or equivalent

B1

- (ii) a and b (two values) unknown
 Impossible to calculate
 Cannot be calculated
Or equivalent

B1

2

[6]

Q12.

- (a) Ordering values gives:
*May be implied by correct median
 or correct IQR*

(a) 14 15 18 20 25 25 26 27 29 32 34 37 37 (b)

Ignore any reference to a and b

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M1

Median = 26

CAO

A1

$IQR = 34 - 18 = 16$

CAO

A2

Special Case:

Identification that $LQ = 18$ and $UQ = 34$

Both CAO

(A1)

4

- (b) (i) Two values (25 and 37) of mode

No unique value
 Sparse data
 Many different values
Or equivalent

B1

(ii) *a* and *b* (two values) unknown
 Impossible to calculate
 Cannot be calculated
Or equivalent

B1

2

(c) Mean = $\frac{\sum x}{n} = \frac{390}{15} = 26$
 CAO

B1

If not identified, assume order is $\bar{x}$ then *s*

SD ($\sum x^2 = 11472$) = 9.4 to 9.8

AWFW (9.423 & 9.754)

Treat rounding of a correct stated answer to an integer as ISW

B1

Special Case:

Evidence of $\frac{\sum x}{15}$

Can only be awarded if no marks scored elsewhere in (c)

(M1)

2

[8]

Q13.

(a) Mean = $\frac{\sum fx}{\sum x} = \frac{247}{52} = 4.75$ or $4\frac{3}{4}$

$$\frac{247}{52} \Rightarrow B1$$

$$CAO \left(4.75 = 5 \Rightarrow ISW \right)$$

$$4 \frac{39}{52} \Rightarrow B2$$

B2

If B0 but evidence of $\frac{\sum fx}{52}$

(M1)

Median (26, 26½) = 5
CAO

B2

Stated identification of 26 or 26½

(B1)

If B0 but evidence of cumulative frequencies

F: (0) 1 3 12 25 32 45 51 52

Need to see attempt at ≥ 4 F-values

(M1)

or

If data assumed continuous so use of

$$4 + \frac{x}{7} \text{ where } 0 < x < 2$$

(4 < median < 4.29)

Mode(s) = 4 and 6

CAO both (so mode = 5 $\Rightarrow$ B0)

B1

5

- (b) Mode(s)
More than one mode/value

CAO

Or equivalent; eg not unique

B1

Two modes/values

No unique mode/value

Dep only on previous B1 scored

B1dep

Notes:

If data treated as two separate sets, then only marks available are B1 B1dep in (b)

If averages confused then mark (a) as stated

eg median = 4 and 6 $\Rightarrow$ B0 in (a) and in

(b) "median, as two values" $\Rightarrow$ B0 B0

Modes = 1 and 13 $\Rightarrow$ B0 in (a) but B1 B1dep available in (b)

2

[7]

Q14.

(a) Mean = $\frac{\sum fx}{\sum f} = \frac{275}{99} = 2.77 \text{ to } 2.78$
AWFW (2.778)

B1

If not identified, assume order is $\bar{x}$ then s

Treat rounding to integers as ISW

SD $(\sum fx^2 = 933) = 1.3(0) \text{ to } 1.32$
AWFW (1.307 & 1.314)

B2

Special Case:

Evidence of $\frac{\sum fx}{99}$

Can award if no marks scored in (ii)

(M1)

3

(b) (i) Mean₁₆₃ = $\frac{99 \times \text{Mean}_{99}}{163}$ or $\frac{\sum fx \text{ from (a)(ii)}}{163}$
Or equivalent; may be implied by an answer within range

M1

= 1.68 to 1.69

AWFW (1.687)

A1

2

(ii) Increase

CAO; or equivalent (1.696)
Ignore any working (1.702)

B1

1

(iii) Data is (positively/negatively) skewed /
 not symmetric / bimodal / not bell-shaped from
 frequency distribution / given table

Or equivalent

B1

or

$[C's \text{ mean in (b)(i)}] - 2 \times [C's \text{ SD in (a) (ii)}] <$
 $(-1.75 \text{ to } -0.90)$

or

$[C's \text{ mean in (b)(i)}] - 2 \times [1.69 \text{ to } 1.71] < 0$
 Thus claim appears **not valid**

Or equivalent

Dependent upon previous B1

B1dep

2

[8]

Q15.

(a) (i) Median (50) = 3

CAO

*Do not award marks if correct
 answers are based on shown
 incorrect method; eg accept use of
 99/2, etc but not 276/2, etc*

B1

If not identified, then assume order is median then IQR

$IQR (75 - 25) = 4 - 2 = 2$

CAO; but

$25^{th} \text{ value} \Rightarrow IQR = 2 \Rightarrow B0$

B2

Special Cases:

Identification that LQ = 2 and UQ = 4

Both CAO

(B1)

Statement of ≥ 4 cumulative frequencies

F: 14 49 74 87 96 98 99

*Can award if no marks scored in (i)
even if then applied to continuous data*

(M1)

3

(ii) Mean = $\frac{\sum fx}{\sum f} = \frac{275}{99} = 2.77 \text{ to } 2.78$
 AFWW (2.778)

B1

If not identified, assume order is $\bar{x}$ then s
Treat rounding to integers as ISW

SD $(\sum fx^2 = 933) = 1.3(0) \text{ to } 1.32$
 AFWW (1.307 & 1.314)

B2

Special Case:

Evidence of $\frac{\sum fx}{99}$
Can award if no marks scored in (ii)

(M1)

3

(b) (i) Mean₁₆₃ = $\frac{99 \times \text{Mean}_{99} \text{ or } \sum fx \text{ from (a) (ii)}}{163}$
*Or equivalent; may be implied by
an answer within range*

M1

= 1.68 to 1.69
 AFWW (1.687)

A1

2

(ii) Increase
 CAO; or equivalent (1.696)
 Ignore any working (1.702)

B1

1

- (iii) Data is (positively/negatively) skewed / not symmetric / bimodal / not bell-shaped from frequency distribution / given table

Or equivalent

B1

or

$$[C's \text{ mean in (b)(i)}] - 2 \times [C's \text{ SD in (a) (ii)}] < 0$$

(-1.75 to -0.90)

or

$$[C's \text{ mean in (b)(i)}] - 2 \times [1.69 \text{ to } 1.71] < 0$$

Thus claim appears **not valid**

Or equivalent

Dependent upon previous B1

B1dep

2

[11]

Q16.

(a) (i)

$x:$	0	1	2	3	4	5	6	7	8	9
$F:$	30	109	208	276	336	360	371	377	379	380

$$\text{Median } (\approx 190.5^{\text{th}}) = 2$$

CAO; B0 if shown method incorrect

B2

$$\text{Interquartile range } (\approx 285.75^{\text{th}} - \approx 95.25^{\text{th}}) = 4 - 1 = 3$$

*CAO; B0 if shown method incorrect
B1 for identification of 4 and 1*

B2

If neither is correct but F attempted

$$\text{Allow for median} = 1 + \frac{x}{99}$$

(M1)

and matched correctly with ≥ 5 x -values

A1

4

(ii) Mean $\left(\bar{x}\right) = 2.56 \text{ to } 2.57$

AWFW (2.56316)

B2

(2.5 to 2.6)

AWFW

$$\sum fx = 974 \text{ and } \sum fx^2 = 3546$$

(B1)

Standard Deviation (s_n, s_{n-1}) = 1.66 to 1.67

AWFW (1.66187)

B2

(1.6 to 1.7)

AWFW (1.66406)

(B1)

If neither is correct but $\sum fx$ attempted

(M1)

and result divided by 380

(M1)

4

(b) Average: Same/similar/greater in 2004/05

OE; dep on 2 and 2.5 to 2.6

B1 dep

Spread: Similar/greater in 2004/05

OE; dep on 3 and 1.6 to 1.7

B1 dep

2

[10]

Q17.

(a) (i)

$x:$	0	1	2	3	4	5	6	7	8	9
$F:$	30	109	208	276	336	360	371	377	379	380

Median ($\approx 190.5^{\text{th}}$) = 2

CAO; B0 if shown method incorrect

B2

Interquartile range ($\approx 285.75^{\text{th}} - \approx 95.25^{\text{th}}$)

$$= 4 - 1 = 3$$

CAO; B0 if shown method incorrect
B1 for identification of 4 and 1

B2

If neither is correct but F attempted

$$\text{Allow for median} = 1 + \frac{x}{99}$$

(M1)

and matched correctly with ≥ 5 x -values

(A1)

4

(ii) Mean ($\bar{x}$) = 2.56 to 2.57
AWFW (2.56316)

B2

(2.5 to 2.6)
AWFW
 $\sum fx = 974$ and $\sum fx^2 = 3546$

(B1)

Standard Deviation (s_n, s_{n-1}) =

1.66 to 1.67

AWFW (1.66187)

B2

(1.6 to 1.7)

AWFW (1.66406)

(B1)

If neither is correct but $\sum fx$ attempted

(M1)

and result divided by 380

(M1)

4

- (b) (i) Average: Same/similar/greater in 2004/05
OE; dep on 2 and 2.5 to 2.6

B1 dep

Spread: Similar/greater in 2004/05
OE; dep on 3 and 1.6 to 1.7

B1 dep

2

- (ii) Rule applies to data that is
 (approximately) symmetric/normal/bell-shaped
OE

B1

Data for 2005/06 is (positively) skewed/
 not symmetric/not normal/not bell-shaped
OE

B1

2

[12]

Q18.

- (a) Ordering:
 0 0 13 28 35 40 47 51 63 77 *a*
May be implied by 40 and/or 63 and 13

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M1

Median (6th) = 40
CAO

B1

$IQR = Q_3(9^{th}) - Q_1(3^{rd})$
 $= 63 - 13$

Identification of 63 and 13

(B1)

= 50

CAO

B2

4

- (b) (i) *Mode:*
 Zero is not representative / sensible reason

Wide range of (known) values
 Small number of values mostly different
Or equivalent

B1

- (ii) *Range:*
 Largest value, a , is unknown
 Cannot be calculated
Or equivalent

B1

2

[6]

Q19.

- (a) Mean ($\bar{x}$) = 39.3 to 39.4
AWFW (39.35)

B1

Standard Deviation (s_n, s_{n-1}) = 12.3 to 12.7
AWFW (12.358 or 12.679)

B2

If **neither** correct **but** working shown, then

Mean $\left(\bar{x}\right) = \frac{\sum x}{20}$

$$\sum x = 787 \quad \sum x^2 = 34023$$

Used

(M1)

3

- (b) Median = 42
CAO

B2

Median = 41.5 or 39 or 40
CAO

(B1)

Interquartile Range = $55 - 31 = 24$
CAO; allow B1 for identification of 31

and 55; B0 if method shown is incorrect

B2

Interquartile Range = 21 to 27

AWFW

(B1)

4

- (c) (i) **Mode:** eg
 Does not exist
 If exists, must be > 60 or 58
 All / too many different values
 Sparse data
 OE

B1

- (ii) **Range:** eg
Maximum value is unknown / > 60 or 58
 OE; accept 'slowest' but not 'smallest'

B1

2

[9]

Q20.

- (a) (i) Mode = 2
 CAO

B1

Range = 15
 CAO

B1

2

- (ii) CF: 4 17 41 58 73 84 89 95
 x: 0 1 2 3 4 9 14 15

Median (48th) = 3

CAO; B0 if shown method is incorrect

B2

Interquartile Range (72nd – 24th)
 = 4 – 2 = 2

CAO

Allow B1 for identification of 4 and 2

B0 if shown method is incorrect

B2

If neither correct but CF attempted
and matched correctly with ≥ 5 x -values

$$\text{Allow for median} = 2 + \frac{x}{17}$$

(M1)(A1)

4

(iii) Mean $(\bar{x}) = 4.2$

$$\text{CAO } \sum fx = 399$$

B2

Standard Deviation (s_n, s_{n-1})

$$\sum fx^2 = 3111$$

$$= 3.88 \text{ to } 3.91$$

$$\text{AWFW (3.887 or 3.907)}$$

B2

If neither correct but mid-points of 7 and 12 seen

(B1)

and use of mean $(\bar{x}) = \frac{\sum fx}{95}$

$$\text{Allow for } 4.1 \leq \bar{x} \leq 4.3$$

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4

- (b) (i) Unknown values (16) have no effect on
median and IQR **or** median and IQR are
exact values but $\bar{x}$ and s are estimates

B1

1

- (ii) Use all available data
or
Enable further analyses

B1

1

[12]

Q21.

$$\sum fx = 8025$$

$$\sum fx^2 = 739975$$

Mean ($\bar{x}$) = 80.2 to 80.3

AWFW 80.25

B2

Standard Deviation (s_n, s_{n-1}) = 30.9 to 31.2

AWFW 30.97882 or 31.13489

B2

MPs (x): 25, 35, 50, 70, 90, 110, 135, 165

At least 4 correct

(B1)

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{100}$$

Use of

(M1)

[4]

Q22.

(a) B(15, 0.3)

use of in (a)

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(i) $P(K = 5) = P(K \leq 5) - P(K \leq 4)$

$$P(K = 5) = \binom{15}{5} (0.3)^5 (0.7)^{10}$$

may be implied

M1

$$= 0.7216 - 0.5155 = 0.2055 \text{ to } 0.2065$$

AWFW (0.2061)

A1

3

- (ii) (Fewer than) half $\Rightarrow$ 7 or $7\frac{1}{2}$ or 8
stated or implied

B1

Thus require $P(K \leq 7 \text{ or } < 8)$
used or implied by correct answer

M1

$$= 0.9495 \text{ to } 0.9505$$

AWFW (0.9500)

A1

3

- (iii) $P(2 < K < 7) =$ 0.8689 or 0.9500

M1

minus 0.1268 or 0.2969

M1

$$= 0.7415 \text{ to } 0.7425$$

AWFW (0.7421)

A1

3

or
 $B(15, 0.3)$ expressions stated for at least 3 terms
 within $2 \leq K \leq 7$

or implied by a correct answer

M1

Answer

A2

- (b) (i) Mean, $\mu = np = 15 \times 0.4 = 6$
CAO

B1

Variance, $\sigma^2 = np(1-p) = 6 \times 0.6 = 3.6$
use of $\sigma^2 = np(1-p)$

M1

Standard deviation = $\sqrt{3.6} = 1.89$ to 1.9
AWFW; or equivalent

A1

3

(ii) Mean, $\bar{x} = 6$

CAO ($\Sigma x = 60$)

CSO if evidence of $np(1-p)$ or 1.9

B1

Standard deviation, s or $\sigma = 2.82$ to 2.99
AWFW; or equivalent. ($\Sigma x^2 = 440$)

B1

2

(iii) Means are same/equal

*ft on 2 means; accept $\frac{6}{15} = 0.4$ if not
 contradicted by $\bar{x}$ in (ii)*

B1ft

Standard deviations are different
dependent on 2 correct SDs

B1 dep

Reason to doubt validity of Kirk's claim
dependent on 2 correct SDs

B1 dep

3

[17]

Q23.

Mean = 3.75

CAO

$\Sigma fx = 150$

B1

1

Standard deviation = 1.84 to 1.87

AWFW

$\Sigma fx^2 = 698$

B2

3

$s_{n-1}^2 = 3.47$ to 3.48

and $s_x^2 = 3.38$ to 3.39

*Substitution of values into correct formula
for variance or SD or*

SD = 3.38 to 3.48 AFWW

M1

1

[3]

Q24.

(a) (i) Mean ($\bar{x}$) = 24.7 to 25.7

AWFW (25.2)

B2

Standard Deviation (s_n, s_{n-1})
= 16.7 to 17.7

AWFW (17.1474 or 17.2338)

B2

MPs (x): 5.5, 15.5, 23, 28, 33, 38, 45.5, 75.5

At least 4 correct

(B1)

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{100}$$

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(M1)

4

(b) Data is skewed or not symmetric

Discrete data or counts

One valid reason

B1

1

(Mean – 2 × SD) < 0 $\Rightarrow$ negative counts

(c) (i) Since sample size large ($n > 30$)
can use Central Limit Theorem

Either point

B1

1

- (ii) Mean = μ
 CAO; not $\bar{x}$ or its value

B1

$$\text{Variance} = \frac{\sigma^2}{100}$$

Accept $\frac{\sigma^2}{n}$ or $\frac{(\text{their SD})^2}{100}$, etc

B1

2

- (d) 99% $\Rightarrow z = 2.57$ to 2.58
 AFWW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$

Use of

Must have ($\div \sqrt{n}$) with $n > 1$

M1

Thus $25.2 \pm 2.5758 \times \frac{17.1 \text{ or } 17.2}{\sqrt{100}}$

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ft on $\bar{x}$, z and $s > 0$; not on n

A1ft

(20.8, 29.6)
 AWR

A1

4

- (e) UCL < 30
 ft on CI

B1ft

so

↑ dep

Reject claim that $\mu > 30$ B1ft
ft on CI

7/100 or 7% of $X > 50$ (from table)
CAO

B1

so

↑ dep

Reject claim that often $X > 50$
CAO

B1

4

[16]

Q25.

(a) Mean x is 155

B1

Variance is $24\,125 - 155^2$

... = 100

M1

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A1

3

(b) $\Sigma x = 1377$ now

M1

So mean height is 153 cm

Allow NMS; condone absence of units throughout

A1

$\Sigma x^2 = 241\,250 - 173^2 (= 211321)$

M1

variance is $\frac{241250 - 173^2}{9} - 153^2$

m1

... $\approx 72 \text{ cm}^2$

Accept AWRT 71

A1

5

[8]

Q26.

Use mid interval values

Condone one error

M1

Correct method for mean

using values from given intervals

M1

Est mean 1.54 (minutes)

NMS 3/3

A1

$\Sigma fx^2 / \Sigma f = 3.53$

seen or implied

B1

Method for SD or variance

using values from given intervals

M1

Est SD 1.08 (minutes)

(3sf needed) NMS 3/3; allow AWRT 1.07 or 1.08

A1

6

[6]

Q27.

(a) $\frac{2160}{40} = 54 \text{ (}^\circ\text{F)}$

B1

$$\sqrt{\frac{116680}{40} - 54^2} = 1 \quad (^\circ\text{F})$$

allow even if $\sqrt{\quad}$ unclear

allow $\sqrt{1}$

M1A1

(b) $12 \frac{2}{9} (^\circ\text{C})$

allow 12.2. Not 12.3

M1A1

$$\frac{5}{9} (^\circ\text{C})$$

allow 0.56

B1f.t.

[6]

Q28.

- (a) Mean mass is 200 grams

Condone absence of units in (a)

B1

Method for variance or SD

Must be used

M1

Variance is 900 g^2

PI; allow 900 given as answer

A1

SD is 30 grams

ft wrong variance

A1F

- (b) Mean cost is £4

ft wrong value in (a)

B1F

2

SD is 60p

*ditto; error in units penalised only once in (b)/(c);
no penalty if mark previously lost*

B1F

(c) Mean of total cost is £5

ft wrong value in (b)

B1F

2

SD is 60p

ditto

B1F

[8]

Q29.

(a) $\bar{x} = 12$ $s = 1.73$

12 cao

1.73(1.73 – 1.735) allow

1.63(1.63 – 1.635)

0.4 cao

B1/B1

2

(b) 0.4

0.4 cao

B1

1

(c) Binomial

B1

1

(d) (i) $P(X > 15) = 0.9029$

*attempted use of $B(30, p)$, their p
method – allow $P(\geq 15)$*

B1/M1

$= 0.0971$

0.0971(0.097 – 0.0972)

A1

3

- (ii) Mean 12
12 cao

B1

$$\text{s.d.} = \sqrt{30 \times 0.4 \times 0.6} = 2.68$$

*Method – allow variance if called variance, their p
2.68(2.68 – 2.69)*

M1/A1

3

- (e) Unsuitable
Unsuitable

E1

standard deviations very different
*ft s.d. different
max 1 if previous M not earned*

E1 ft

2

(Mean same but this would be the case no matter what the distribution of X)

- (f) P not constant – some houses more likely to put boxes out than others
P not constant/not independent

E1

1

[13]

Q30.

- (a) (i) Binomial $n = 10$ $p = 0.4$

B1

$$P(3 \text{ or fewer}) = 0.382$$

*Correct use of tables or correct method of calculation
0.382-0.383*

M1 A1

3

- (ii) $(> 4) = 1 - (4 \text{ or fewer}) = 1 - 0.6331$
 *$(> 4) = 1 - P(4 \text{ or fewer})$ or
correct method of calculation*

M1

$$= 0.367$$

$$0.366 - 0.368$$

A1
2

(b) Mean = $10 \times 0.4 = 4$
cao

B1

Standard deviation = $\sqrt{10 \times 0.4 \times 0.6} =$
Method for s.d. or variance (if called variance)

M1

1.55
 $1.54 - 1.56$

A1
3

(c) 3.92
 $3.92 - 3.93$

3.17 (allow 3.05)
 $3.17 - 3.18 (3.045 - 3.055)$

B1

B1
2

- (d) Mean similar but standard deviation much greater than model
Standard deviations different

E1

Binomial not good model
Binomial not suitable model

E1

Siballi's beliefs not plausible
Siballi's beliefs not supported

E1
3

[13]

Q31.

(a)

Class mid - point	Frequency
45	12
65	54
85	68
105	41
135	23

Method for mid-points

M1 B2 B2

5

87.1 (87.0 – 87.1)

87.2 24.0 (23.9 – 24.1)

$x = 87.1$ $s = 24.0$

Allow M1 A1 if method shown

(b) Journeys from Essex similar average time/ Essex slightly shorter

E1

Essex more variable

E1

2

[7]